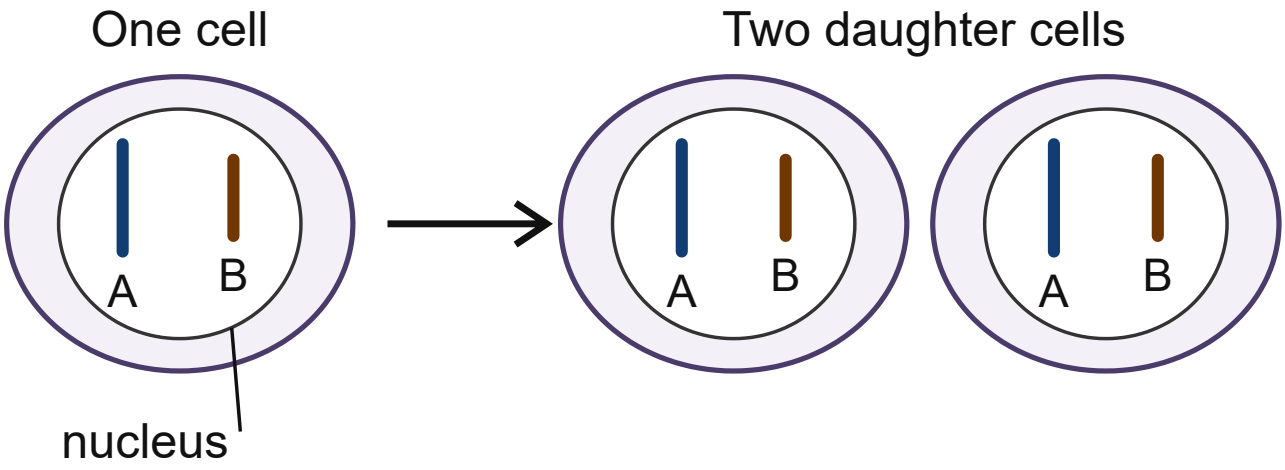
Reference A matching chromosome sets



Compare the full A and B set in both daughter cells. Each new cell has a matching set.

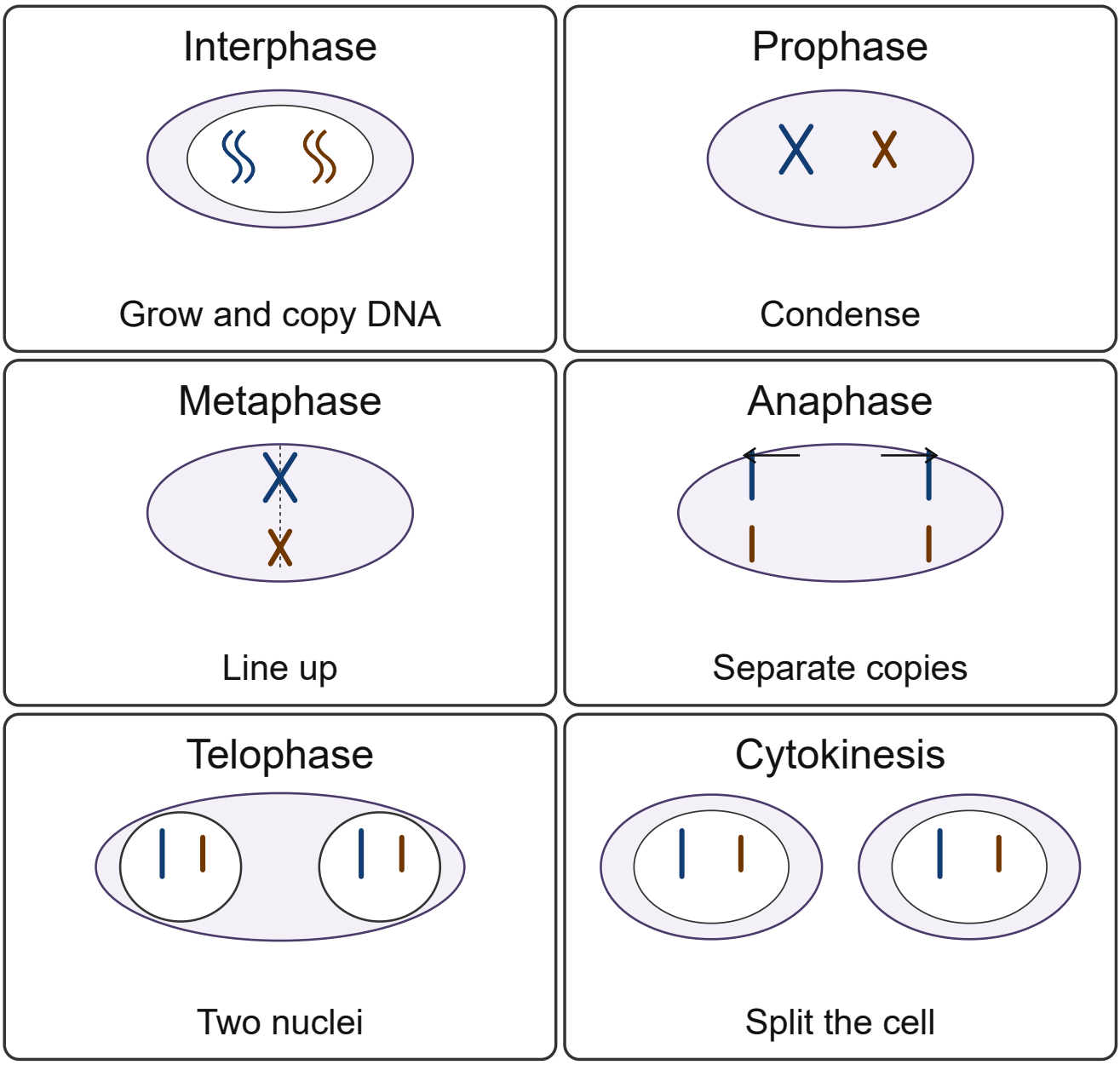
Reference only. Use labels as well as colours. The symbols are simplified chromosome types, not a full picture of real DNA.

Step	What changes
Interphase	The cell grows and copies its DNA.
Mitosis	Copied chromosomes separate into two nuclei.
Cytokinesis	The cell splits into two daughter cells.

In the number case, six chromosomes in the starting body cell means six in each daughter after ordinary mitosis and cell division. A fresh case starting with eight gives eight in each daughter.

Made by Matt

Reference B the six named stages



Interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Four middle phases belong to mitosis.

Reference only. Interphase comes before mitosis. Cytokinesis follows the formation of the two nuclei. The four named mitosis phases are prophase, metaphase, anaphase and telophase.

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Science · LAUNCH · Autumn 2 · Week 1 · SCI-LAUNCH-A2-W01-L1

Arrival and exit Supported

Use the labelled picture on Reference A for question 1. Respond in your usual mode.

1. Use the picture only. How many cells are shown after the final split: one or two?

2. Which picture is first: the one cell on the left or the two cells on the right?

3. Point to the labelled nucleus in the first cell.

4. Show one change from the first picture to the last.

Exit ticket

1. Put these three cards in order: interphase, mitosis, cytokinesis.

2. Do the new body cells have matching sets or half the original set?

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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You do Supported

Use your book or this sheet. Use Reference A and B. One response per task.

1. Order interphase, mitosis and cytokinesis. Show A and B in each new cell. [3]
Staff I ☐ S ☐ C ☐

Use the three cards beside the model. First the DNA is copied; then the sets separate; last the cell splits.

---

2. Finish: Each new cell matches because DNA was ___ and the copied chromosomes ___ into two nuclei. [3]
Staff I ☐ S ☐ C ☐

Use copied and separated. Point to a full matching set in each new nucleus.

---

3. How do copy, separate and split work together when new cells are needed for repair? Show it or give one short explanation. [3]
Staff I ☐ S ☐ C ☐

Start with copying. Check that each new cell has a set before you describe repair.

Feedback and response · Supported

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

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